

5G Reference Signal-based Positioning in a Reinforcement Learning Approach

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Abstract—The demand for enhanced positioning in complex environments has intensified with the integration of 5G New Radio (NR). Existing algorithms suffer from Non-Line-of-Sight (NLOS) degradation, high signaling overhead, and reduced multipath accuracy, hindering practical deployment. This paper proposes a reinforcement learning-based positioning method leveraging 5G reference signals. The algorithm separates localization into angle estimation (via uplink/downlink feedback) and distance measurement using a Weighted Average Reference Signal Received Power (WA-RSRP) technique to mitigate noise. The Twin Delayed Deep Deterministic Policy Gradient (TD3) framework optimizes the accuracy-overhead tradeoff. Simulation results in practical dynamic scenarios demonstrate that: (1) The proposed algorithm reduces error by approximately 70.77% compared to benchmarks at the same signaling overhead level; (2) With a 95% reduction in signaling overhead, the algorithm still achieves around 30% lower error than baseline approaches.

Index Terms—Signaling Overhead, 5G Positioning, Reinforcement Learning, 5G Reference Signal.

I. INTRODUCTION

Nowadays, positioning is closely associated with the daily life of people, supporting various applications in autonomous drivings, smart factory, urban canyons and many other fields [1], [2]. However, existing traditional positioning algorithms primarily rely on channel feature extraction to extract angular and distance information from the channel characteristics [3]. Despite their applicability, these conventional algorithms tend to show a sharp decline in performance at low signal-to-noise ratios (SNR), which is largely due to noise interference. Then, a clock offset compensation method is proposed in [4] to achieve stationary user equipment (UE) localization within 10 meters under the context of line-of-sight (LOS). However, the scenarios of localization are not confined to LOS environments as non-line-of-sight (NLOS) conditions are included [5].

As highlighted in [6], Artificial Intelligence (AI) models can be used to significantly facilitate crucial localization tasks, such as LOS/NLOS identification and time-angle measurements. Following this trend, machine learning techniques are explored in [7], [8], including neural networks, random forests, and deep learning networks, so as to estimate UE location based on signal strength measurements. While AI-driven methods produce an improved performance. For high-

precision localization, the challenge must be addressed that signal strength measurements are susceptible to environmental interference. At present, fingerprinting localization has been proposed as a viable solution to this issue, with [9] demonstrating the effectiveness of fingerprinting algorithms in achieving robust positioning in dynamic environments. Nevertheless, a considerable rise in their application costs results from the dependency on a comprehensive fingerprint database.

In this paper, we propose a 5G reference signal-based positioning algorithm in a reinforcement learning approach. The main contributions are as follows:

- 1) The proposed integrated positioning framework, which synergies bidirectional angle estimation with RSRP-based weighted average, exhibits 70.77% lower average positioning error than benchmark algorithms while maintaining comparable signaling overhead.
- 2) The proposed positioning framework incorporates Twin Delayed Deep Deterministic Policy Gradient (TD3) reinforcement learning to minimize signaling overhead while maintaining positioning accuracy. Experimental validation demonstrates that with 95% less signaling overhead, the system achieves approximately 30% higher positioning precision compared to existing benchmarks.
- 3) Unlike existing methods limited to the Indoor Factory (InF) scenario, the proposed algorithm demonstrates better positioning accuracy in all three environments compared with existing benchmarks: InF, Urban Micro-cell (UMi), and Urban Macro-cell (UMa), demonstrating its robustness and adaptability to varying deployment conditions.

II. SYSTEM MODEL

To address the issues of low positioning accuracy in complex environments and high signaling overhead in practical positioning processes, we propose a positioning algorithm based on TD3 and RSRP weighted averaging. Fig. 1 illustrates the flow chart of our proposed positioning algorithm, which includes three main components: position system based on the uplink and downlink feedback, TD3-based 5G localization algorithm and three dimensions (3D) position results determination.

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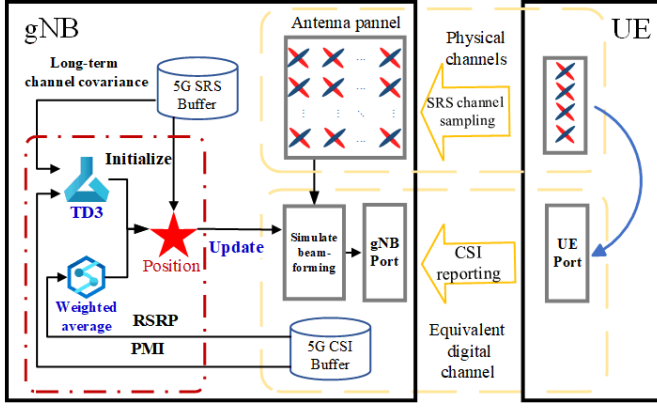


Fig. 1. The flowchart of our proposed positioning algorithm

A. Positioning system based on uplink and downlink feedback

The Next Generation NodeB (gNB) can obtain channel information using uplink sounding reference signal (SRS) measurements and downlink channel state information reference signal (CSI-RS) feedback. Thereby reducing the impact of multipath loss on the positioning algorithm and improving its positioning accuracy in real-world complex environments. Then, the following sections will elaborate on three aspects: angle extraction based on SRS, angle optimization based on Precoding Matrix Indicator (PMI), and distance extraction based on PMI.

1) *Angle extraction based on SRS*: The gNB can configure the UE to transmit SRS for uplink channel measurements periodically. Also, the gNB extracts angular information from the covariance matrix of the sampled channel matrix $\hat{H}$. The vectors $\mathbf{h}_{t,p1}, \mathbf{h}_{t,p2} \in \mathbb{C}^{N_k \times N_u \times M_B}$ represent the decomposition of $\hat{H}$, yielding different polarized subchannels $(p1, p2)$ at time t , where N_u and N_k represent the number of UE antennas and subbands, respectively. Additionally, M_B represents the number of gNB antennas. Corresponding to different polarization directions, the instantaneous channel covariance matrices are denoted as $\mathbf{R}_{t,p1}$ and $\mathbf{R}_{t,p2}$, respectively.

$$\mathbf{R}_{t,p1} = \mathbb{E}(\mathbf{h}_{t,p1}^* \mathbf{h}_{t,p1}), \mathbf{R}_{t,p2} = \mathbb{E}(\mathbf{h}_{t,p2}^* \mathbf{h}_{t,p2}) \quad (1)$$

where $\mathbb{E}$ represents the expectation operator. $*$ superscript denotes conjugate. The gNB obtains the **instantaneous channel covariance** $\mathbf{R}_t$ at time t by summing the weighted contributions from different polarization directions.

$$\mathbf{R}_t = \frac{\text{tr}(\mathbf{R}_{t,p1}) \mathbf{R}_{t,p1} + \text{tr}(\mathbf{R}_{t,p2}) \mathbf{R}_{t,p2}}{\text{tr}(\mathbf{R}_{t,p1}) + \text{tr}(\mathbf{R}_{t,p2})} \quad (2)$$

where tr is the trace operator of a matrix. Then, the **eigenvector** $\mathbf{B}_{t,i}$ of the i -th subpath at time t can be obtained.

Assume that the best estimate angle for the i -th subpath is $(\hat{\phi}_i, \hat{\theta}_i)$. The beamforming vector corresponding to the estimated angle is denoted as $\hat{\mathbf{B}}_{t,i}(\hat{\phi}_i, \hat{\theta}_i)$. $\hat{\mathbf{B}}_{t,i}$ is supposed to exhibits the maximal correlation with $\mathbf{B}_{t,i}^*$. It is expressed as Eq.3:

$$(\hat{\phi}_i, \hat{\theta}_i) = \underset{(\hat{\phi}_i, \hat{\theta}_i)}{\text{argmax}} \text{tr}(\mathbf{B}_{t,i}^* \hat{\mathbf{B}}_{t,i}(\hat{\phi}_i, \hat{\theta}_i)) \quad (3)$$

Since the subpath introduces noise and random perturbations to the LOS path, **multiple subpath angles are averaged**

to mitigate these effects. Then, the azimuth estimate $\bar{\phi}$ and the elevation estimate $\bar{\theta}$ can be obtained for the LOS path.

$$\bar{\phi} = \frac{1}{N_k N_u} \sum_{i=0}^{N_k N_u} \hat{\phi}_i, \bar{\theta} = \frac{1}{N_k N_u} \sum_{i=0}^{N_k N_u} \hat{\theta}_i \quad (4)$$

However, due to the **sparsity** of the channel and the **sampling characteristics** of SRS, the R_t extracts pertains to N_{CH} subpaths, potentially causing significant deviation in the angular measurement from $(\bar{\phi}, \bar{\theta})$. Therefore, it is necessary to acquire more subpath information through **multiple measurements** performed **within the period T** . The optimal estimated angles of UE are denoted as $(\hat{\phi}, \hat{\theta})$. The beamforming vector corresponding to $(\hat{\phi}, \hat{\theta})$ is denoted as $\mathbf{B}(\hat{\phi}, \hat{\theta})$. Also, $\mathbf{B}(\hat{\phi}, \hat{\theta})$ is supposed to exhibit the maximum correlation with the eigenvectors of all subpaths within the period T , which is expressed as follows:

$$(\hat{\phi}, \hat{\theta}) = \underset{(\hat{\phi}, \hat{\theta})}{\text{argmax}} \sum_{t=0}^T \sum_{i=0}^{N_{CH}} \text{tr}(\mathbf{B}_{t,i}^* \mathbf{B}(\hat{\phi}, \hat{\theta})) \quad (5)$$

Then, the **long-term accumulated** covariance matrix $\bar{\mathbf{R}}_t$ is presented as follows:

$$\bar{\mathbf{R}}_t = \chi \bar{\mathbf{R}}_{t-1} + (1 - \chi) \mathbf{R}_t, \quad (6)$$

where χ denotes the attenuation coefficient. Finally, by SVD on $\bar{\mathbf{R}}_t$ and extracting the angular $(\bar{\phi}, \bar{\theta})$ and $(\hat{\phi}, \hat{\theta})$.

2) *Angle optimization based on PMI*: This paper introduces a correction algorithm based on the PMI to mitigate errors caused by frequent SRS measurements. Initially, the analog beam direction $\mathbf{a}$ [10], channel matrix $\mathbf{H}$ and the corresponding analog precoding matrix $\mathbf{F}_{RF}$ are determined. With $\mathbf{F}_{RF}$ applied to channel transmission, the resulting **equivalent digital channel** is denoted as $\mathbf{H}_{eff} = \mathbf{H} \mathbf{F}_{RF}$. The UE then reports the reference signal to transmit the information about the equivalent digital channel.

Afterwards, the UE cycles through all PMI matrices, selects the one that maximizes the equivalent energy with the covariance matrix, and reports it to the gNB. As described in [11], the **codebook** $\mathbf{W}$ corresponding to the current configuration is derived. For the k -th subband, where i_k denotes the optimal codebook index, the optimal digital precoding matrix is represented by $\mathbf{F}_{BB}[k] = \mathbf{W}_{i_k}$.

Therefore, the received signal $\mathbf{y}[k]$ of k -th subband can be expressed as follows:

$$\mathbf{y}[k] = \mathbf{H}_{eff} \mathbf{W}_{i_k} \mathbf{x}[k] + \mathbf{n}[k], \quad (7)$$

where $\mathbf{x}[k]$ denotes the transmitted symbols of the k -th subcarrier; $\mathbf{n}[k] \in \mathbb{C}^{N_r}$ denotes a Gaussian noise vector characterized by a zero mean and a variance of $\frac{N_0}{2}$ per real dimension. By maximizing the correlation energy between $\mathbf{H}_{eff}$ and $\mathbf{W}_{i_k}$, i_k can be expressed as follows:

$$i_k = \underset{i_k}{\text{argmax}} \text{tr}((\mathbf{H}_{eff}[k] \mathbf{W}_{i_k})^* \mathbf{H}_{eff}[k] \mathbf{W}_{i_k}) \quad (8)$$

$k = 1, \dots, K$ where i_k is reported to the gNB through CSI-RS feedback. The gNB selects the corresponding digital precoding matrix based on i_k , as described below.

where $\mathbf{W}_{i_k}$ represents the optimal digital precoding matrix. As $\mathbf{F}[k]$ enhances the correlation with $\mathbf{H}$ based on $\mathbf{F}_{\text{RF}}$, instantaneous angles can be extracted more accurately.

Afterwards, the optimal angle estimation is performed following Eq.10.

$$(\hat{\phi}, \hat{\theta}) = \underset{(\hat{\phi}, \hat{\theta})}{\operatorname{argmax}} \operatorname{Tr} \left((\mathbf{B}(\hat{\phi}, \hat{\theta}) \mathbf{F}[k])^* (\mathbf{B}(\hat{\phi}, \hat{\theta}) \mathbf{F}[k]) \right) \quad (10)$$

3) *Distance extraction based on RSRP*: As the distance d between the UE and gNB correlates with the RSRP, it is proposed in this study to estimate the distance d based on the RSRP feedback from the UE. The estimation is performed using a **path loss model**. RSRP measurement P is expressed as Eq.11:

$$P = c - 10\beta \log_{10}(d) - n_0, \quad (11)$$

where c is a constant, d is the distance between the gNB and the UE, and n_0 is the zero-mean noise related to combining small-scale and large-scale fading random variables. Also, the Direct Average RSRP(DA-RSRP) algorithm can be used to reduce the measurement error caused by n_0 . Measured over a period T , the average RSRP $\bar{P}_t$ is expressed as follows:

$$\bar{P}_t = \frac{1}{T} \sum_{i=1}^T P_i = c - 10\beta \log_{10}(\bar{d}_t) - \bar{n}_0, \quad (12)$$

where $\bar{d}_t$ represents the estimated distance at time t obtained using DA-RSRP; P_i represents the RSRP from the i -th measurement; $\bar{n}_0$ represents the residual noise from multiple measurements.

In the scenarios involving **UE mobility**, the impact of the error $\bar{n}_0$ becomes more significant. For this reason, a weighted averaging method is applied in this study to minimize the errors caused by motion for RSRP, instead of direct averaging.

The **weighted RSRP** $P_w(t)$ at time t can be expressed as follows:

$$P_w(t) = P_w(t-1)w + P_s(t)(1-w), \quad (13)$$

where $P_s(t)$ represents the instantaneous RSRP value at the current time t , with $P_w(0)$ initialized as $P_s(0)$. High-speed users are more trusting of the current RSRP value, showing preference for a weighting coefficient closer to 0. Conversely, low-speed users prefer a weighting coefficient closer to 1. Substituting $P_w(t)$ into Eq.12 yields the RSRP-weighted distance estimate d_t^* at time t .

B. Reinforcement learning based 5G localization algorithm

This paper proposes a TD3-based reinforcement learning framework for high-precision position estimation with minimal signaling overhead in complex environments. This strategic choice stems from TD3's dual advantages over conventional methods: Compared to Deep Deterministic Policy Gradient (DDPG), TD3 enhances robustness through target policy smoothing and dual critics, effectively suppressing NLOS noise and environmental dynamics; conversely, unlike Proximal Policy Optimization (PPO)'s stochastic policy averaging, TD3's deterministic gradient optimization directly maximizes Q-function extremum, thereby improving estimation accuracy.

The state is defined as shown in Eq.14 to effectively leverage the long-term and short-term channel characteristics within the reinforcement learning framework.

$$s = (\phi_{\text{up}}, \theta_{\text{up}}, \phi_{\text{down},k}, \theta_{\text{down},k}) \quad k = 1, \dots, K, \quad (14)$$

where ϕ_{up} and θ_{up} represent the angles reflecting the long-term channel characteristics; $\phi_{\text{down},k}$ and $\theta_{\text{down},k}$ represent the angle information that reflects the instantaneous channel characteristics at k -th subband.

Herein, the angular domain is discretized into 360 actions to achieve an angular resolution of 1° . Based on the conventional wireless network, the maximum values are set for azimuth and elevation angles as $\phi_{\text{max}} = 60^\circ$ and $\theta_{\text{max}} = 60^\circ$, respectively. Thus, the complete action domain A can be defined as follows:

$$A = (\phi_{\text{norm}}, \theta_{\text{norm}}) \quad \phi_{\text{norm}}, \theta_{\text{norm}} \in (-1, 1), \quad (15)$$

where ϕ_{norm} and θ_{norm} represent the action value for the azimuth angle and the elevation angle, respectively. For the UE, angle estimation is achieved by multiplying the action value by the maximum angle value (ϕ^*, θ^*) .

$$\phi^* = \phi_{\text{norm}} \phi_{\text{max}}, \theta^* = \theta_{\text{norm}} \theta_{\text{max}} \quad (16)$$

The correlation between the current positioning direction and long-term channel characteristics is taken as the reward to mitigate the impact of UE channel conditions on the reward. This reward is expressed as follows:

$$r = \operatorname{tr}(\mathbf{a}^*(\phi, \theta) \mathbf{U}) \quad (17)$$

where $\mathbf{a}^*(\phi, \theta)$ denotes the analog beam associated with the spatial direction (ϕ, θ) . The vector $\mathbf{U}$ represents the principal eigenvector extracted from the covariance matrix $\bar{\mathbf{R}}_t$.

C. Three-dimensional positioning results

The positions of the gNB and UE are expressed as $\mathbf{p}_b = [p_b^x, p_b^y, p_b^z]$ and $\mathbf{p}_u = [p_u^x, p_u^y, p_u^z]^T$, respectively. Assume that there are l subpaths, with the 0-th path being the LOS path. The angle parameters of the LOS path can be obtained from Eq.18.

$$\begin{aligned} \phi_{\text{Tx},0} &= \pi + \phi_{\text{Rx},0} = \arccos \left(\frac{p_u^x - p_b^x}{\sqrt{(p_u^x - p_b^x)^2 + (p_u^y - p_b^y)^2}} \right) \\ \theta_{\text{Tx},0} &= -\theta_{\text{Rx},0} = \arcsin \left(\frac{p_u^z - p_b^z}{\sqrt{(p_u^x - p_b^x)^2 + (p_u^y - p_b^y)^2 + (p_u^z - p_b^z)^2}} \right) \end{aligned} \quad (18)$$

where $\phi_{\text{Tx},0}$ and $\phi_{\text{Rx},0}$ represent the azimuth angles of transmission and reception for the 0-th path, respectively; $\theta_{\text{Tx},0}$ and $\theta_{\text{Rx},0}$ denote the elevation angles of transmission and reception for the 0-th path, respectively. According to the Section II-A3 and Section II-B, (ϕ^*, θ^*, d^*) are given, the estimate position of the UE $\mathbb{P}_u = [\mathbb{P}_u^x, \mathbb{P}_u^y, \mathbb{P}_u^z]^T$ can be calculated according to Eq. 19.

$$\begin{aligned} \mathbb{P}_u^x &= d^* \cos(\theta^*) \cos(\phi^*) + p_b^x \\ \mathbb{P}_u^y &= d^* \cos(\theta^*) \sin(\phi^*) + p_b^y \\ \mathbb{P}_u^z &= d^* \sin(\theta^*) + p_b^z \end{aligned} \quad (19)$$

For brevity, technical details are omitted here and provided in [12].

III. PERFORMANCE EVALUATION

A. Simulation Settings

Using the industry standard QUAsi Deterministic RadIo channel GenerAtor (QuADRIGa) [13] for realistic simulations,

TABLE I
CHANNEL SIMULATION PARAMETERS

Parameter	Value
UE Height	1.5m
Outdoor gNB Height	[8m,10m,25m]
Number of antennas	32
Antenna spacing	5.36mm
Antenna polarization	$\pm 45^\circ$
Number of subband	17
Subcarrier spacing	30KHz
Carrier frequency	28GHz
Number of UE	10
Simulation time	2s
Speed	3km/h
Measurement/feedback cycle	20ms
Uplink-to-downlink ratio	2:3
SRS frequency configuration	comb2
SRS/CSIRS Port	4
CSI-CDM-type	FD-CDM2
CSIRS (k,l)	(k0,l0), (k0+2,l0)
CSIRS CDM index group	0,1
SRS first OFDMsymbolinTimeDomain	13

our algorithm achieves better reliability and practicality than existing methods through environment-based validation. This paper aims primarily to investigate UE positioning in three scenarios: InF, UMi, and UMa. The configuration follows the specifications outlined in [11]. Table I presents the channel simulation parameters. Position derivation is computed per reference signal cycle, enabling millisecond-level location calculation. Then, the calculation steps for overhead percentages are detailed below.

Step 1: Calculate the total resources: The uplink resources are given by Eq. 20:

$$17^1 \times 40^2 \times 14^3 \times \frac{2}{5}^4 = 3808 \quad (20)$$

The downlink resources are given by Eq.21:

$$17 \times 40 \times 14 \times \frac{3}{5}^5 = 5712 \quad (21)$$

Step 2: Calculate the uplink resource consumption:

The user's frequency domain allocation spans the entire bandwidth based on the fixed configuration. SRS resource consumption is given by Eq.22:

$$17 \times 4^6 \times \left(\frac{10}{2}\right)^7 = 340 \quad (22)$$

Step3: Calculate the downlink resource consumption

With FD-CDM2 selected, CSIRS resource consumption is expressed as:

$$17 \times 1^8 \times \frac{10}{2} = 85 \quad (23)$$

Step 4: Determine the proportion of reference signal overhead in both the uplink and downlink.

¹17 represents the count of subbands.

²40 represents the number of slots within 20ms at a 30kHz subcarrier spacing.

³14 is the number of OFDM symbols within a single slot.

⁴ $\frac{2}{5}$ represents the proportion of total resources allocated to the uplink.

⁵ $\frac{3}{5}$ represents the proportion of total resources allocated to the downlink.

⁶4 represents the number of symbols to calculate the SRS.

⁷10 is the number of UEs, 2 is the number of band multiplexes and

¹⁰ $\frac{10}{2}$ represents the number of UEs supported in the time domain

⁸1 represents the number of symbols to calculate the CSIRS.

The signaling overhead proportion for uplink SRS can be expressed as:

$$\frac{17 \times 4 \times 5}{17 \times 40 \times 14 \times (2/5)} = 8.93\% \quad (24)$$

The signaling overhead proportion for downlink CSIRS can be expressed as:

$$\frac{17 \times 1 \times 5}{17 \times 40 \times 14 \times (3/5)} = 1.49\% \quad (25)$$

Uplink SRS and downlink CSI-RS signaling overheads are 8.93% and 1.49% at 20ms reporting. By adjusting the interval to 20xbase RP, all algorithms (including benchmarks) reduced overhead uniformly to 5% of baseline (1RP), ensuring consistent cross-parameter comparisons.

B. 3D positioning simulation results

In this section, the proposed algorithm is compared against the Multiple Signal Classification (MUSIC) [3], Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) [14] and PPO [7] algorithms under various dynamic scenarios, at varying levels of signaling overhead.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS ACROSS SCENARIOS

Scenario	Algorithm	Overhead	10% error (m)	50% error (m)	90% error (m)
INF	TD3+WA-RSRP	1RP	0.50	1.5	2.50
		20RP	3.54	5.05	11.11
	PPO+WA-RSRP	1RP	1.50	3.00	4.50
		20RP	4.04	6.06	17.68
	MUSIC+DC-RSRP	1RP	7.58	12.12	16.67
	ESPRIT+DC-RSRP	1RP	5.56	10.10	15.15
UMI	TD3+WA-RSRP	1RP	0.50	1.50	3.00
		20RP	3.03	5.56	16.16
	PPO+WA-RSRP	1RP	1.00	2.00	4.00
		20RP	5.56	10.61	18.18
	MUSIC+DC-RSRP	1RP	13.63	17.67	24.74
	ESPRIT+DC-RSRP	1RP	11.61	15.15	21.71
UMA	TD3+WA-RSRP	1RP	1.00	2.50	4.00
		20RP	3.03	7.07	18.18
	PPO+WA-RSRP	1RP	1.50	3.50	6.50
		20RP	2.53	5.05	18.69
	MUSIC+DC-RSRP	1RP	14.14	19.69	23.73
	ESPRIT+DC-RSRP	1RP	11.61	17.17	22.22

To compare the positioning performance of high and low overhead in different positioning algorithms, the cumulative distribution function(CDF) are used as the metrics of statistics. Table II summarizes the positioning results of different algorithms under various scenarios. As clearly indicated in the table, in the case of 1 RP, the 90% error of the PPO+WA-RSRP, MUSIC+DC-RSRP, and ESPRIT+DC-RSRP algorithms in the INF scenario are 4.5, 16.67, and 15.15 m, respectively. However, the corresponding error of the proposed algorithm is 2.5 m, that is, a decrease of approximately 44.44%, 85%, and 83.5%, respectively. Similarly,

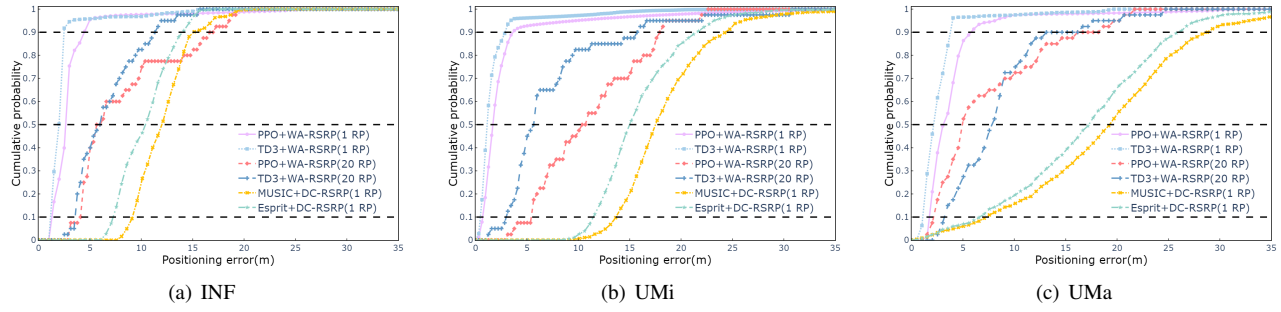


Fig. 2. CDF of positioning errors under different reporting periods (RP)

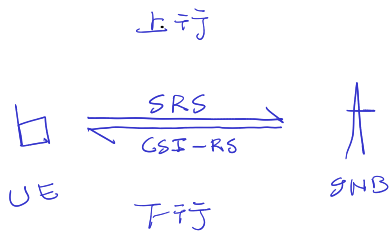
in UMi and UMa scenarios, the 90% error of the proposed algorithm are reduced by an average 67.11% compared with the other three positioning algorithms. In the case of 20 RP, that is, the signaling overhead is reduced by approximately 95%, the 90% error of the proposed algorithm in the INF scenario is 11.11 m, and the corresponding error of the PPO+WA-RSRP is 17.68 m. This indicates that, under the same low signaling overhead, our proposed algorithm still reduces the error by approximately 37.16% compared to the PPO algorithm. Moreover, the proposed algorithm maintains approximately 33.35% and 26.67% lower error margin at 20 RPs compared to the MUSIC and ESPRIT algorithms operating at 1 RP, respectively. Fig. 2 illustrates the CDF of the positioning errors of different algorithms under different RP. Fig. 2(a) clearly demonstrates that the proposed algorithm achieves significant error reduction compared to other algorithms in INF under both low and high signaling overhead conditions. Similarly, as evidenced by Fig. 2(b) and (c), our algorithm maintains superior performance in both UMi and UMa environments.

IV. CONCLUSION

This study presents a novel positioning algorithm that simultaneously enhances localization accuracy and optimizes spectral efficiency, offering practical implementation potential for 5G New Radio (NR) networks. The proposed integrated positioning framework synergizes bidirectional angle estimation with RSRP-based weighted averaging while incorporating a TD3 reinforcement learning mechanism to minimize signaling overhead without compromising positioning accuracy. Simulation results reveal three key advancements: (1) In the INF scenario, the proposed method achieves a remarkable 70.98% reduction in 90% error compared to conventional approaches under equivalent signaling overhead; (2) Signaling overhead is reduced by 95% while maintaining equivalent positioning precision; (3) Cross-environment robustness is demonstrated through validation in urban UMi, UMa, and INF scenarios. Notably, the algorithm's real-time capability is ensured through co-implementation with 5G network functions on a unified computing platform. Future research will focus on millimeter-wave array enhancement through optimized antenna array configurations and radiation pattern synthesis, potentially enabling sub-meter positioning accuracy in 3GPP-compliant networks.

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SRS: gNB 获得上行信道矩阵 $\hat{H}$
 CSI-RS: UE 测量后, 计算下行信道状态, 再反馈给 gNB.
 (该文中只是 PMI)

1. SRS 提取上行信道估计 UE $\xrightarrow{\text{SRS}}$ gNB

gNB 端: $\hat{H} \rightarrow$ 瞬时极化 R_{t,P_1}, R_{t,P_2} (双极化) $\xrightarrow{\text{相加}}$ $R_t \xrightarrow{\text{SVD}}$ 特征值 $\frac{\omega}{2} B_{t,i}$

$$(\hat{\phi}_i, \hat{\theta}_i) = \arg \max \text{tr} (B_{t,i}^* \hat{B}_{t,i}(\hat{\phi}_i, \hat{\theta}_i))$$

长期统计协方差矩阵

↓ 对天线维、子载波维取平均 (消除噪声、多径、扰动影响)

$$\bar{R}_t = \chi \bar{R}_{t-1} + (1-\chi) R_t$$

$$\bar{\Phi} = \frac{1}{N_k N_0} \sum_i \hat{\phi}_i, \quad \bar{\Theta} = \frac{1}{N_k N_0} \sum_i \hat{\theta}_i$$

↓ 累计算多次测量 (信道稀疏性, SRS 采样时延)

$$(\hat{\Phi}, \hat{\Theta}) = \arg \max \sum_{t=0}^T \sum_{i=0}^{N_{ch}} \text{tr} (B_{t,i}^* \hat{B}_{t,i}(\hat{\Phi}, \hat{\Theta}))$$

2. 基于 PMI 的角度估计

PMI, precoding matrix indicator

UE 端: $H_{eff} = H F_{RF}$, $i_k = \arg \max_{i_k} \text{tr} ((H_{eff}[k] W_{i_k})^* H_{eff}[k] W_{i_k})$

↓ 反馈

gNB 端: 根据 i_k 找到 W_{i_k} , $F[k] = F_{RF} W_{i_k}$

$$(\hat{\Phi}, \hat{\Theta}) = \arg \max \text{tr} ((B(\hat{\Phi}, \hat{\Theta}) F[k])^* (B(\hat{\Phi}, \hat{\Theta}) F[k]))$$

3. 基于 RSRP 的距离估计

$$P = c - 10 \beta \lg(\Delta) - n_p$$

$$\left\{ \begin{array}{l} \text{Direct Average: } P_t = \frac{1}{T} \sum_{t=1}^T P_i \\ \text{Weighted Averages: } P_w(t) = w P_w(t-1) + (1-w) P_s(t) \end{array} \right.$$

4. 强化学习框架

状态: $(\phi_{up}, \theta_{up}, \phi_{down,k}, \theta_{down,k})$, $k=1, \dots, K$

动作: $A = (\phi_{norm}, \theta_{norm})$, $\phi_{norm}, \theta_{norm} \in (-1, 1)$

$$\phi^* = \phi_{norm} \times 60^\circ$$

$$\theta^* = \theta_{norm} \times 60^\circ$$

奖励: $r = \text{tr}(A^*(\phi, \theta), U)$

总结: 本人在英特尔研究院 (当时还在触科, 现在全搬到朝阳环球了) 的 NEM 部门实习了 3 个月的项目。由于与华科合作到期, 公司内部仅存着初期的代码, 需要观察了结果继续做研究。总的来讲, 我认为华科圈不再成长, 但通过该项目, 理解了企业内如何做通信研究。总的来讲是不需要做监督学习 (即不需要打标签的数据)。负责该项目的 (即作师兄) 与本人一开始的 leader 都于我实习期间相继被裁, 令人唏嘘。